

# 19ELMOA2

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## Problem

Find all functions  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  such that for all surjective functions  $g : \mathbb{Z} \rightarrow \mathbb{Z}$ , the function  $f + g$  is also surjective.

(A function  $g$  is surjective over  $\mathbb{Z}$  if for all integers  $y$ , there exists an integer  $x$  such that  $g(x) = y$ .)

**Solution.** Define  $s_0, s_1, s_2, \dots$  such that

$$g(s_0) = 0, \quad g(s_1) = 1, \quad g(s_2) = -1, \quad g(s_3) = 2, \dots$$

and so on.

Say

$$B = \{b_0, b_1, b_2, \dots\}$$

is the set such that

$$g(b_i) = -f(b_i),$$

where

$$B \subseteq \{s_0, s_1, \dots\} = S.$$

We have the following cases:

- **$B$  is finite.**

Let

$$B = \{b_0, b_1, b_2, \dots, b_k\}.$$

Define  $g' : \mathbb{Z} \rightarrow \mathbb{Z}$  such that

$$g'(n) = \begin{cases} b_{i+1}, & \text{if } n = b_i, \ 0 \leq i < k, \\ b_0, & \text{if } n = b_k, \\ g(n), & \text{otherwise.} \end{cases}$$

Then  $g'$  is surjective as well, but  $g' + f$  is not surjective. Contradiction.

- **$B$  is infinite.**

Define  $g' : \mathbb{Z} \rightarrow \mathbb{Z}$  such that

$$g'(n) = \begin{cases} b_{2i+1}, & \text{if } n = b_{2i}, i \geq 0, \\ b_{2i}, & \text{if } n = b_{2i+1}, i \geq 0, \\ g(n), & \text{otherwise.} \end{cases}$$

Then  $g'$  is surjective as well, but  $g' + f$  is not surjective. Contradiction.

As a result, for any  $g : \mathbb{Z} \rightarrow \mathbb{Z}$ , exactly one  $b$  satisfies

$$g(s_b) = -f(s_b).$$

Suppose there exist  $m, n \in \mathbb{Z}$  such that

$$f(m) \neq f(n).$$

Let

$$s_m = m, \quad s_n = n.$$

**Claim —** There exists a surjective function  $g$  such that

$$g(s_m) = -f(s_m).$$

*Proof.* Take any surjective function  $h$ , and add a sufficiently large constant  $C$  so that

$$g = h + C$$

satisfies the required condition. Since  $h$  is surjective,  $h + C$  is also surjective.  $\square$

Define  $g' : \mathbb{Z} \rightarrow \mathbb{Z}$  such that

$$g'(k) = \begin{cases} g(s_n), & \text{if } k = s_m, \\ g(s_m), & \text{if } k = s_n, \\ g(k), & \text{otherwise.} \end{cases}$$

Then  $g'$  is surjective as well, but  $g' + f$  is not surjective. Contradiction.

Therefore,

$$f(m) = f(n)$$

for all  $m, n \in \mathbb{Z}$ . Hence  $f$  is constant.

Finally, all constant functions satisfy the condition, so the answer is all constant functions.